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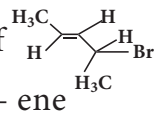
- Haloalkane undergoes nucleophilic substitution and elimination reactions. Primary alkyl halides undergo S_N2 mechanism. If the reactant is chiral, the product formed exhibits inversion of stereo chemical configuration Tertiary alkyl halide undergoes S_N1 mechanism, via carbonium ion formation. If the reactant is chiral, the product formed is optically inactive due to racemisation.
- Organo metallic compound
- Haloalkane reacts with metal to form organometallic compounds like Grignard reagent. It is represented as Rδ- - Mgδ+X. Grignard reagent reacts with variety of substances to give almost all class of organic compounds like alcohols, aldehydes, ketones, acids etc.
- Haloarenes
- Haloarenes are prepared from arenes or by decomposition of benzene diazonium chloride. Haloarenes are more stable than haloalkane. C – X bond in halo arenes is short and strong.
- Under normal conditions halo arenes do not undergo nucleophilic substitution but takes part in electrophilic substitution. Electron withdrawing inductive effect of halogen atom deactivates the benzene ring whereas resonating structure control o, p directing nature of halo arenes.
- Poly halogen compounds
- Organic compounds having two or more halogen atoms are called poly halogen compounds. These compounds are useful in our day to day life but pose environmental threat.
- Chloroform is used as an anasthetic, but because of its toxic nature it has been replaced by less toxic and safer anaesthetic like ethers.
- Iodoform is used as an antiseptic, due to the liberation of free iodine. But it has been replaced by other formulation containing iodine, due to its objectional smell.
- Carbon tetrachloride is used in fire extinguishers. Freons are used as refrigerant. But both these compounds lead to adverse environmental effect.
- DDT is used an effective insecticide. Now a days it is banned because of it's long term toxic effect.

Evaluation

I. Objective type questions (MCQ)

1. The IUPAC name of  is
   a) 2-Bromo pent – 3 – ene
   b) 4-Bromo pent – 2 – ene
   c) 2-Bromo pent – 4 – ene
   d) 4-Bromo pent – 1 – ene

2. Of the following compounds, which has the highest boiling point?
   a) n-Butyl chloride
   b) Isobutyl chloride
   c) t-Butyl chloride
   d) n-propyl chloride